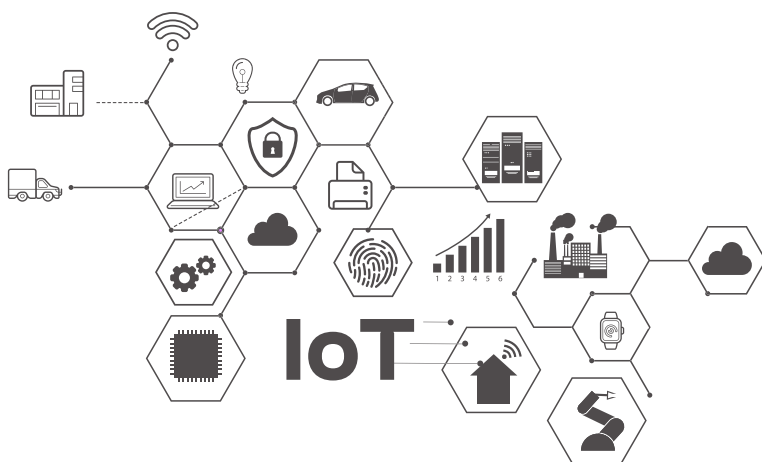


Why Open Source Tools are Popular for Developing an IoT Ecosystem

The technology ecosystem of IoT is complex and includes hardware, connectivity, platforms, middleware, analytics, operating systems, applications and services. Open source technologies are useful in such a scenario.



Enterprises adopting proprietary IoT platforms are now looking at open source as an alternative, recognising the value of an open source community as a hub for innovation and continuous development for IoT. As per IDC, there will be 41.6 billion connected IoT devices, or ‘things’, generating 79.4 zettabytes (ZB) of data in 2025. Sixty-four per cent of the participants in the IDC Global IoT Decision Maker Survey think open source is extremely important in the deployment of IoT. And as per W3.org, 91 per cent of IoT developers use open source software, open hardware or open data in at least one part of their development stack.

We believe that in the coming years, the IoT market will move towards an open source and open standards ecosystem that enables collaborative innovation,

and promotes interoperability, flexibility, and modularity.

Drivers for open source adoption in IoT systems

1. **Cost:** Adoption of the open source technology IoT framework does not involve any money, as it is free to use.
2. **Efficiency:** Adoption of open source helps developers reduce the development time, and hence the cost.
3. **Scalability:** Helps in affordable scaling — billions of devices can be supported by using open source technology.
4. **Innovation:** Building newer applications using open source is permissionless and low risk, which leads to innovation and agility.
5. **Open source API:** Use of open source APIs for the IoT framework offers a common gateway for different software, hardware

and systems to communicate with one another.

6. **Libraries:** An open source IoT framework offers a wide range of libraries, SDKs and open source hardware like Raspberry Pi and Arduino, ensuring that companies remain on the cutting-edge of technology by using different open sourced tools to customise IoT platforms.
7. **Security:** Open source software can protect the data of individuals by implementing really strong encryption (SSH, SSL, PGP, etc).
8. **Interoperability:** Adoption of open source helps in collaboration and interoperability.

Open source IoT ecosystem

Enterprises need to build a modular and flexible IoT solution that is open, interoperable and cost-effective. The open source solution should deliver end-to-end enterprise level IoT needs.

A complete IoT ecosystem requires the following.

- *Hardware:* This mainly covers sensors and devices.
- *Connectivity:* This is needed for transmitting data to the cloud and receiving commands from it.
- *Software:* Helps in analysing the data collected from the sensors and in decision-making.
- *User interface:* This is required to address user interactions.